

Name: _____

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Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. [11 points] Let $f(x) = -\frac{1}{10} (2x^2) (x^2 - 3x)$. Note that we have the following:

$$f'(x) = -\frac{1}{10} (2x^2) (4x - 9) \quad \text{and} \quad f''(x) = -\frac{1}{10} (12x) (2x - 3)$$

You must show your work for all parts to receive credit!

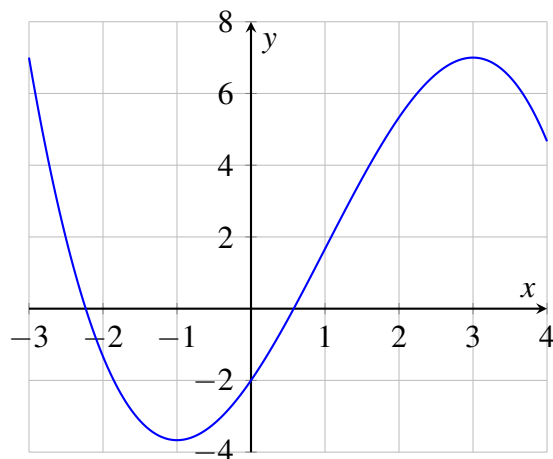
- a. Determine the intervals where f is increasing and where f is decreasing.
- b. Find the x -values where any local maxima occur and where any local minima occur. If none exist, state "none".
- c. Find the intervals where f is concave up and where f is concave down.
- d. Find the x -values where any inflection points occur. If none exist, state "none".

2. [8 points] Evaluate the following limits. **Show your work!**

a. $\lim_{x \rightarrow -\infty} \frac{5x^3 - 7x^5}{8x^5 + 10x^4}$

b. $\lim_{x \rightarrow -\infty} \frac{3(x^2 + 8x)}{\sqrt{x^4 + 2x^2}}$

3. [6 points] Based on the graph of the function $g(x)$ below, indicated whether each of the given values is **positive**, **negative**, **zero**, or **undefined**.



a. $g'(-1)$

d. $g''(-1)$

b. $g'(0.5)$

e. $g''(0.5)$

c. $g'(3)$

f. $g''(3)$